Soft reinforcement learning

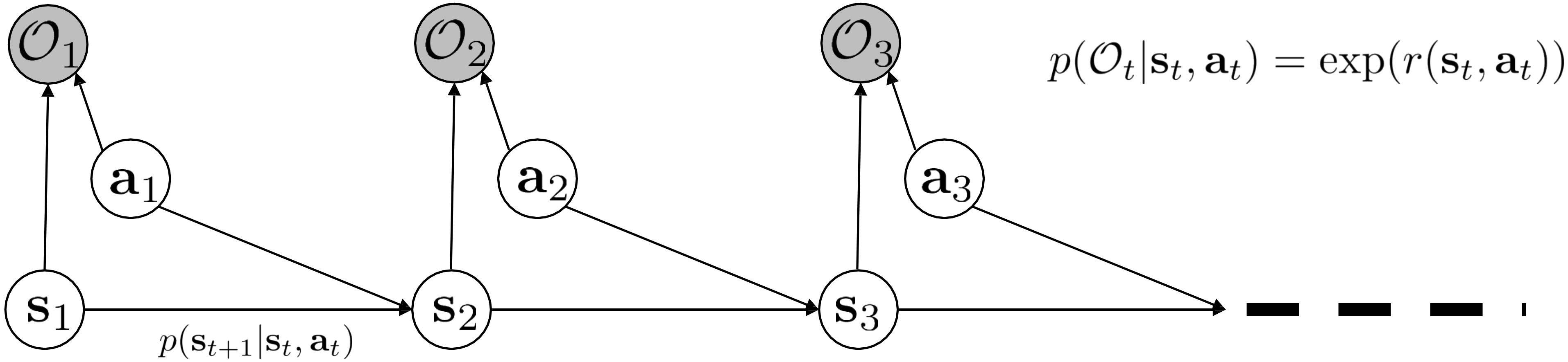
Applications of Deep Learning (69158)

Rubén Martínez Cantín
Dpto. Informática e Ingeniería de Sistemas.

Should agents be ALWAYS optimal?

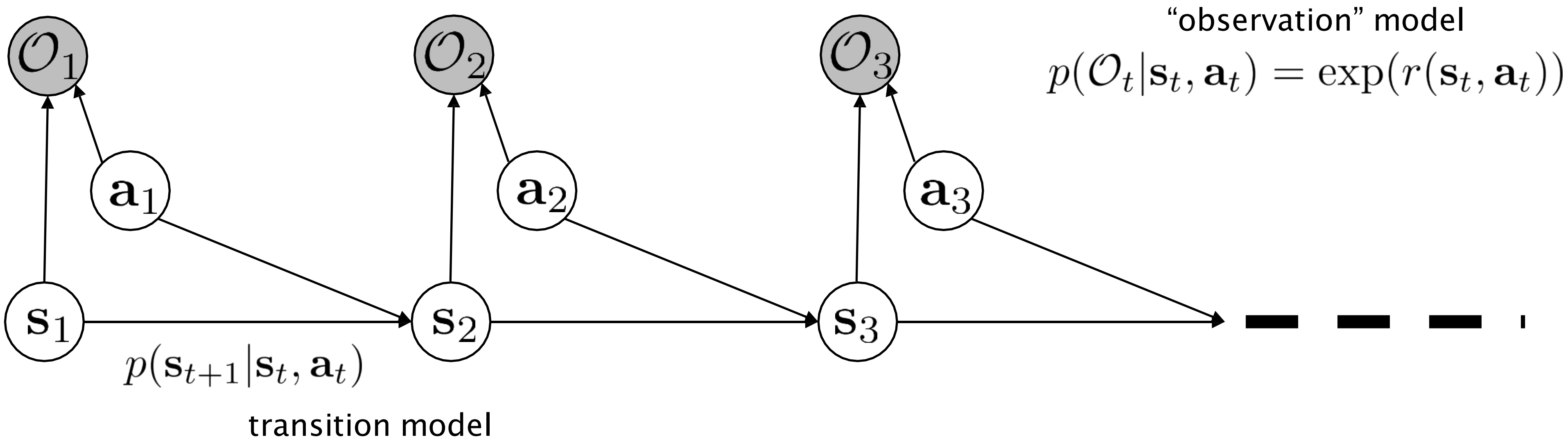
- Human behavior is not optimal
  - Stochastic.
  - Errors and mistakes.
  - Good behavior should still be the most probable.

- Let's build a model without using optimality (max)



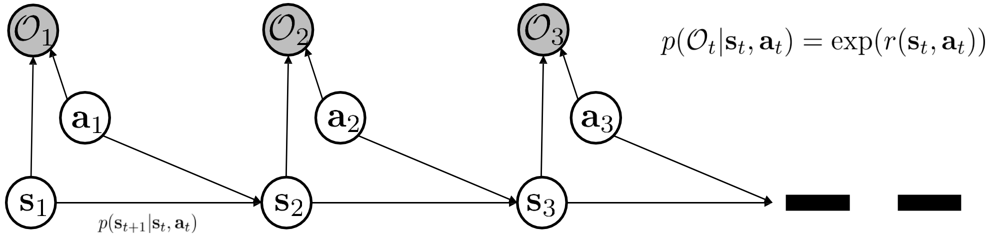
Planning/control as inference

- Can model suboptimal behavior (important for inverse RL and imitation)
- Provides an explanation for why stochastic behavior might be preferred (useful for exploration and transfer learning)
- Can apply inference algorithms to solve control and planning problems (e.g.: message passing or variational inference)



Solving the smoothing problem

- Planning = inference
  - **Smoothing** problem $p(a_t, s_t | \mathcal{O}_{1:T})$
  - Similar to Kalman **filtering** $p(a_t, s_t | \mathcal{O}_{1:t})$



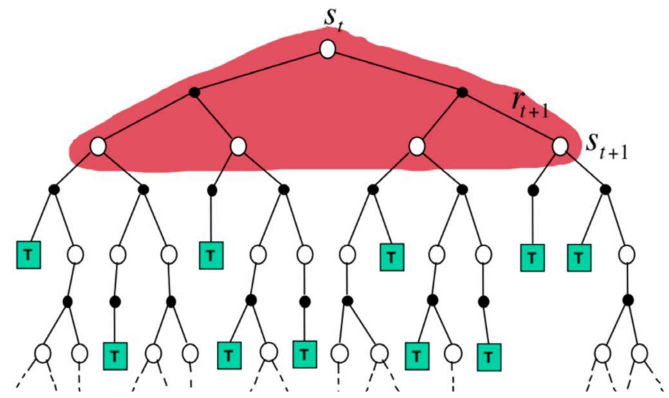
- Exact solution: message passing

1. compute backward messages $\beta_t(\mathbf{s}_t, \mathbf{a}_t) = p(\mathcal{O}_{t:T} | \mathbf{s}_t, \mathbf{a}_t)$
2. compute policy $p(\mathbf{a}_t | \mathbf{s}_t, \mathcal{O}_{1:T})$
3. compute forward messages $\alpha_t(\mathbf{s}_t) = p(\mathbf{s}_t | \mathcal{O}_{1:t-1})$

- Similar to value iteration
  - Backward messages analogous to value computation
  - Forward messages not needed for policy, but useful in some applications (e.g.: inverse RL)

Value iteration remainder

$$V(x_t) \leftarrow \mathbb{E}_\pi [R_{t+1} + \gamma V(x_{t+1})]$$



$$V_{k+1}(x) = \max_a \sum_{x' \in \mathcal{X}} p(x'|x, a) \left(R(x, a, x') + \gamma V_k(x')\right)$$

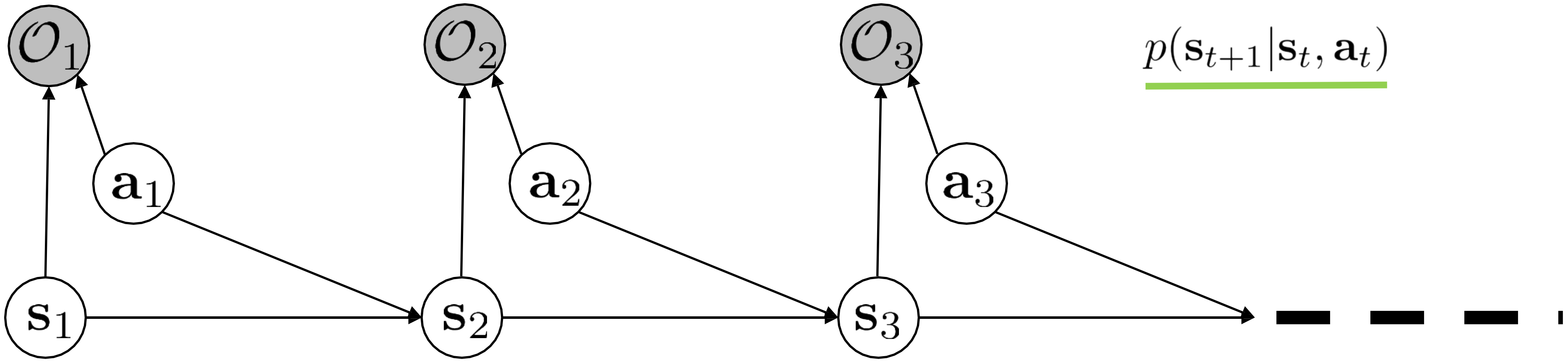
$$\pi(x) \leftarrow \arg\max_{a \in \mathcal{A}} \sum_{x' \in \mathcal{X}} p(x'|x, a)(R(x, a, x') + \gamma V_k(x'))$$

Universidad Zaragoza 1542

Backward pass

$$p(\mathcal{O}_t|\mathbf{s}_t,\mathbf{a}_t) \propto \exp(r(\mathbf{s}_t,\mathbf{a}_t))$$

$$p(\mathbf{s}_{t+1}|\mathbf{s}_t,\mathbf{a}_t)$$



$$\begin{aligned}\beta_t(\mathbf{s}_t,\mathbf{a}_t) &= p(\mathcal{O}_{t:T}|\mathbf{s}_t,\mathbf{a}_t)\\ &= \int p(\mathcal{O}_{t:T},\mathbf{s}_{t+1}|\mathbf{s}_t,\mathbf{a}_t)d\mathbf{s}_{t+1}\\ &= \int p(\mathcal{O}_{t+1:T}|\mathbf{s}_{t+1})p(\mathbf{s}_{t+1}|\mathbf{s}_t,\mathbf{a}_t)p(\mathcal{O}_t|\mathbf{s}_t,\mathbf{a}_t)d\mathbf{s}_{t+1}\end{aligned}$$

$$p(\mathcal{O}_{t+1:T}|\mathbf{s}_{t+1}) = \int \underbrace{p(\mathcal{O}_{t+1:T}|\mathbf{s}_{t+1},\mathbf{a}_{t+1})}_{\beta_t(\mathbf{s}_{t+1},\mathbf{a}_{t+1})}\cancel{p(\mathbf{a}_{t+1}|\mathbf{s}_{t+1})}d\mathbf{a}_{t+1}$$

which actions are likely *a priori*
(assume uniform*)

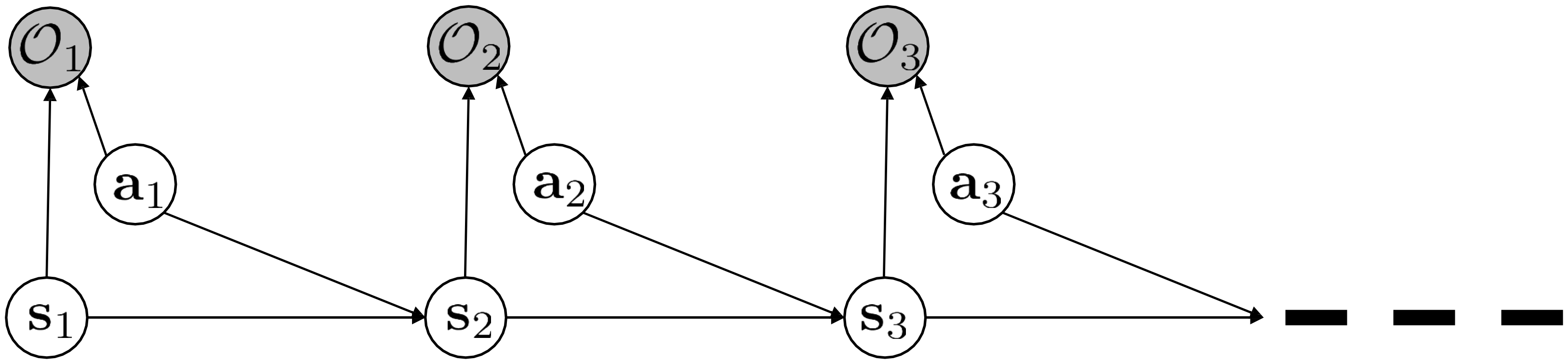
for $t = T-1$ to 1:

$$\beta_t(\mathbf{s}_t,\mathbf{a}_t) = p(\mathcal{O}_t|\mathbf{s}_t,\mathbf{a}_t)E_{\mathbf{s}_{t+1}\sim p(\mathbf{s}_{t+1}|\mathbf{s}_t,\mathbf{a}_t)}[\beta_{t+1}(\mathbf{s}_{t+1})]$$

$$\beta_t(\mathbf{s}_t) = E_{\mathbf{a}_t\sim p(\mathbf{a}_t|\mathbf{s}_t)}[\beta_t(\mathbf{s}_t,\mathbf{a}_t)]$$

*passive dynamics: non-uniform a priori actions can be incorporated in the "observation"

Backward pass summary



$$\beta_t(\mathbf{s}_t, \mathbf{a}_t) = p(\mathcal{O}_{t:T} | \mathbf{s}_t, \mathbf{a}_t)$$

probability that we can be optimal at steps t through T given that we take action $\mathbf{a}_t$ in state $\mathbf{s}_t$

for $t = T - 1$ to 1:

$$\beta_t(\mathbf{s}_t, \mathbf{a}_t) = p(\mathcal{O}_t | \mathbf{s}_t, \mathbf{a}_t) E_{\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)} [\beta_{t+1}(\mathbf{s}_{t+1})]$$

$$\beta_t(\mathbf{s}_t) = E_{\mathbf{a}_t \sim p(\mathbf{a}_t | \mathbf{s}_t)} [\beta_t(\mathbf{s}_t, \mathbf{a}_t)]$$

value iteration algorithm:

1. set $Q(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E[V(\mathbf{s}')]$
2. set $V(\mathbf{s}) \leftarrow \max_{\mathbf{a}} Q(\mathbf{s}, \mathbf{a})$

Backward pass summary

for $t = T-1$ to 1:

$$\beta_t(\mathbf{s}_t, \mathbf{a}_t) = p(\mathcal{O}_t|\mathbf{s}_t, \mathbf{a}_t) E_{\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)}[\beta_{t+1}(\mathbf{s}_{t+1})]$$

$$\beta_t(\mathbf{s}_t) = E_{\mathbf{a}_t \sim p(\mathbf{a}_t|\mathbf{s}_t)}[\beta_t(\mathbf{s}_t, \mathbf{a}_t)]$$

value iteration algorithm:

1. set $Q(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E[V(\mathbf{s}')]$
2. set $V(\mathbf{s}) \leftarrow \max_{\mathbf{a}} Q(\mathbf{s}, \mathbf{a})$

let $V_t(\mathbf{s}_t) = \log \beta_t(\mathbf{s}_t)$

let $Q_t(\mathbf{s}_t, \mathbf{a}_t) = \log \beta_t(\mathbf{s}_t, \mathbf{a}_t)$

$$Q_t(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \overbrace{\log E}^{\text{“soft-max” transition (problem!)}}[\exp(V_{t+1}(\mathbf{s}_{t+1}))]$$

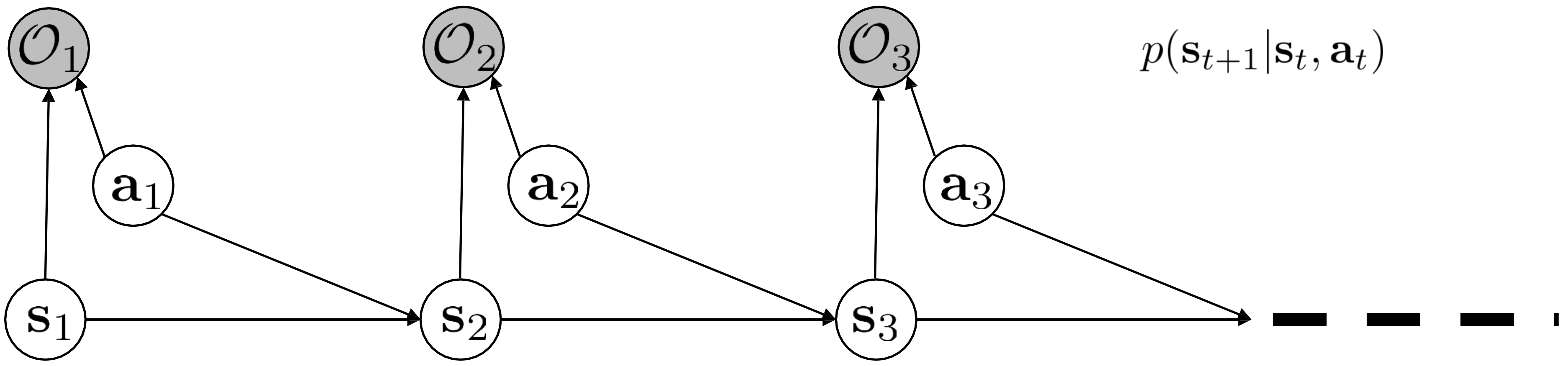
$$V_t(\mathbf{s}_t) = \underbrace{\log \int}_{\text{“soft-max” action (great!)}} \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t)) d\mathbf{a}_t$$

$$V_t(\mathbf{s}_t) \to \max_{\mathbf{a}_t} Q_t(\mathbf{s}_t, \mathbf{a}_t) \text{ as } Q_t(\mathbf{s}_t, \mathbf{a}_t) \text{ gets bigger!}$$

Policy computation

$$p(\mathcal{O}_t|\mathbf{s}_t, \mathbf{a}_t) \propto \exp(r(\mathbf{s}_t, \mathbf{a}_t))$$

$$p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$$



2. compute policy $p(\mathbf{a}_t|\mathbf{s}_t, \mathcal{O}_{1:T})$

$$\beta_t(\mathbf{s}_t, \mathbf{a}_t) = p(\mathcal{O}_{t:T}|\mathbf{s}_t, \mathbf{a}_t)$$

$$\beta_t(\mathbf{s}_t) = p(\mathcal{O}_{t:T}|\mathbf{s}_t)$$

$$\begin{aligned}
p(\mathbf{a}_t|\mathbf{s}_t, \mathcal{O}_{1:T}) &= \pi(\mathbf{a}_t|\mathbf{s}_t) \\
&= \frac{p(\mathbf{a}_t, \mathbf{s}_t|\mathcal{O}_{t:T})}{p(\mathbf{s}_t|\mathcal{O}_{t:T})} \\
&= \frac{p(\mathcal{O}_{t:T}|\mathbf{a}_t, \mathbf{s}_t)p(\mathbf{a}_t, \mathbf{s}_t)/\cancel{p(\mathcal{O}_{t:T})}}{p(\mathcal{O}_{t:T}|\mathbf{s}_t)p(\mathbf{s}_t)/\cancel{p(\mathcal{O}_{t:T})}} \\
&= \frac{p(\mathcal{O}_{t:T}|\mathbf{a}_t, \mathbf{s}_t)}{p(\mathcal{O}_{t:T}|\mathbf{s}_t)} \frac{p(\mathbf{a}_t, \mathbf{s}_t)}{p(\mathbf{s}_t)} = \frac{\beta_t(\mathbf{s}_t, \mathbf{a}_t)}{\beta_t(\mathbf{s}_t)} \cancel{p(\mathbf{a}_t|\mathbf{s}_t)}
\end{aligned}$$

$$\boxed{\pi(\mathbf{a}_t|\mathbf{s}_t) = \frac{\beta_t(\mathbf{s}_t, \mathbf{a}_t)}{\beta_t(\mathbf{s}_t)}}$$

Policy computation with value functions

for $t = T-1$ to 1:

$$Q_t(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \log E[\exp(V_{t+1}(\mathbf{s}_{t+1}))]$$

$$V_t(\mathbf{s}_t) = \log \int \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t))\mathbf{a}_t$$

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \frac{\beta_t(\mathbf{s}_t, \mathbf{a}_t)}{\beta_t(\mathbf{s}_t)}$$

$$V_t(\mathbf{s}_t) = \log \beta_t(\mathbf{s}_t)$$

$$Q_t(\mathbf{s}_t, \mathbf{a}_t) = \log \beta_t(\mathbf{s}_t, \mathbf{a}_t)$$

$$\boxed{\pi(\mathbf{a}_t|\mathbf{s}_t) = \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t) - V_t(\mathbf{s}_t)) = \exp(A_t(\mathbf{s}_t, \mathbf{a}_t))}$$

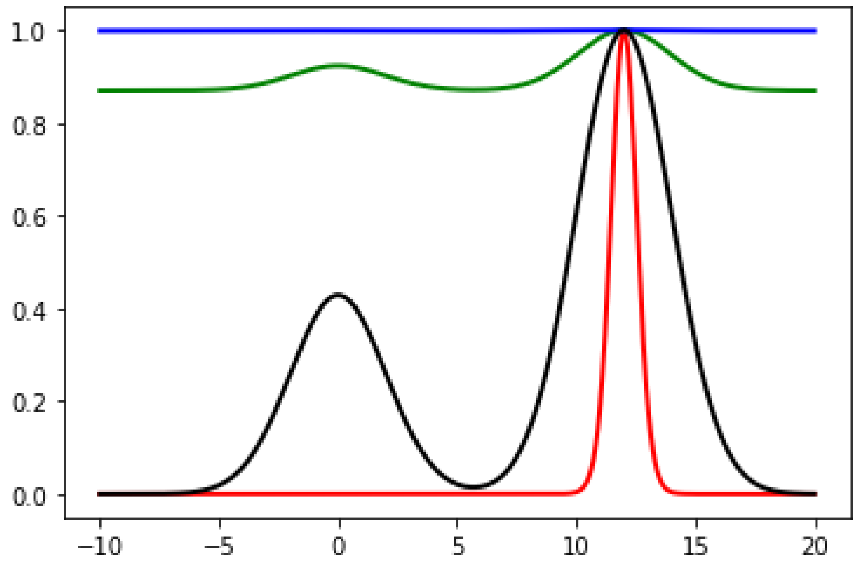
Policy computation summary

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t) - V_t(\mathbf{s}_t)) = \exp(A_t(\mathbf{s}_t, \mathbf{a}_t))$$

$$\text{with temperature: } \pi(\mathbf{a}_t|\mathbf{s}_t) = \exp(\tfrac{1}{\alpha}Q_t(\mathbf{s}_t, \mathbf{a}_t) - \tfrac{1}{\alpha}V_t(\mathbf{s}_t)) = \exp(\tfrac{1}{\alpha}A_t(\mathbf{s}_t, \mathbf{a}_t))$$

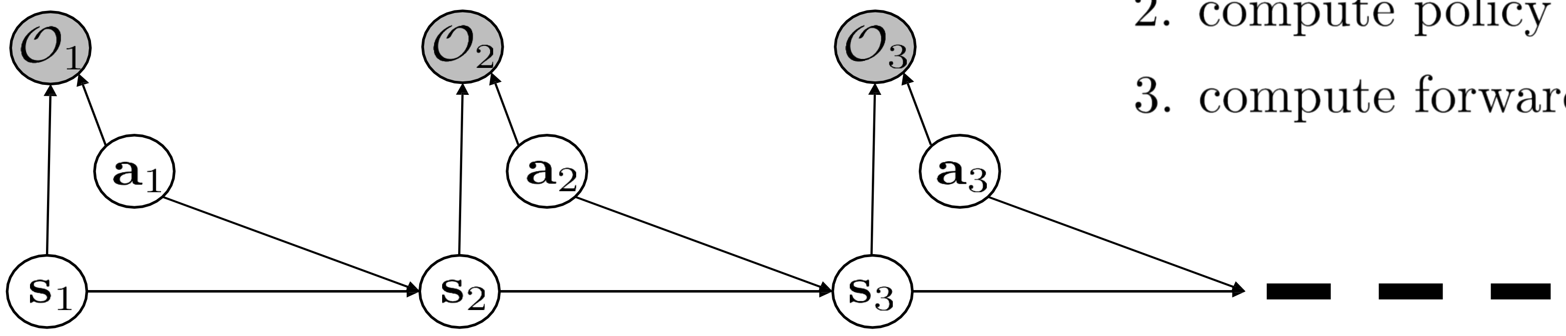
- Natural interpretation: better actions are more probable
- Random tie-breaking
- Analogous to Boltzmann exploration
- Approaches greedy policy as temperature decreases

Effect of temperature α = (0.01, 1, 100).
Black advantage function (normalized plots)



Forward/backward message intersection

1. compute backward messages $\beta_t(\mathbf{s}_t, \mathbf{a}_t) = p(\mathcal{O}_{t:T}|\mathbf{s}_t, \mathbf{a}_t)$
2. compute policy $p(\mathbf{a}_t|\mathbf{s}_t, \mathcal{O}_{1:T})$
3. compute forward messages $\alpha_t(\mathbf{s}_t) = p(\mathbf{s}_t|\mathcal{O}_{1:t-1})$

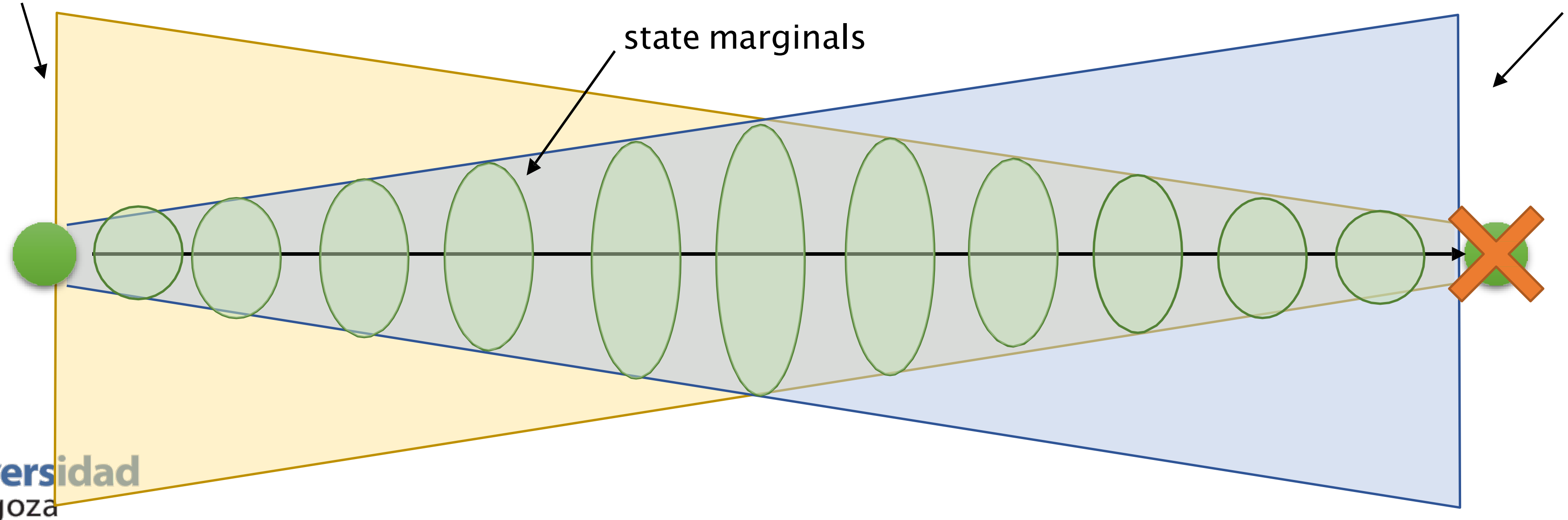


states with high probability of reaching goal

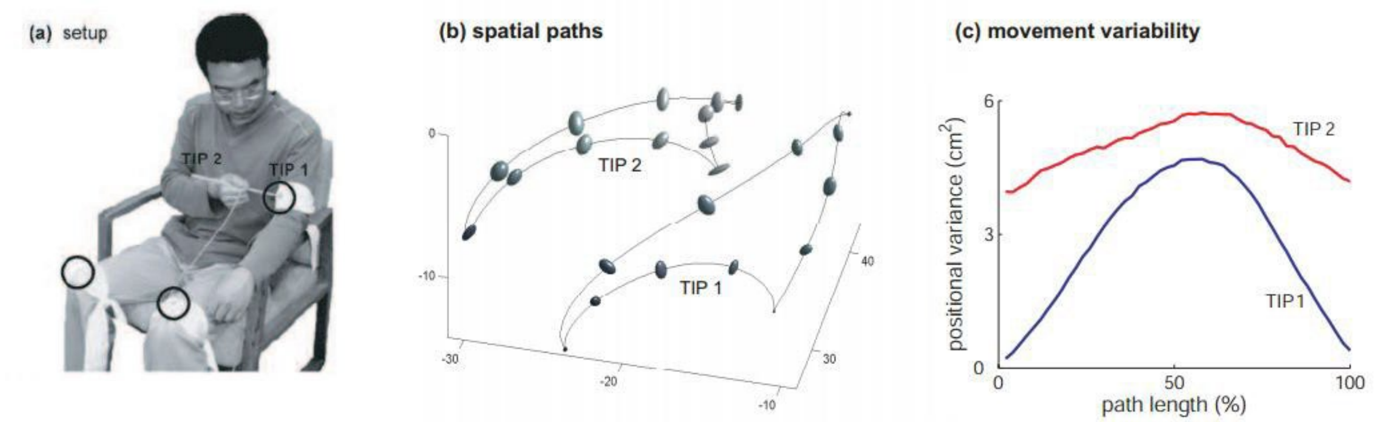
$$p(\mathbf{s}_t) \propto \beta_t(\mathbf{s}_t)\alpha_t(\mathbf{s}_t)$$

states with high probability of being reached from initial state (with high reward)

state marginals



Forward/backward message intersection

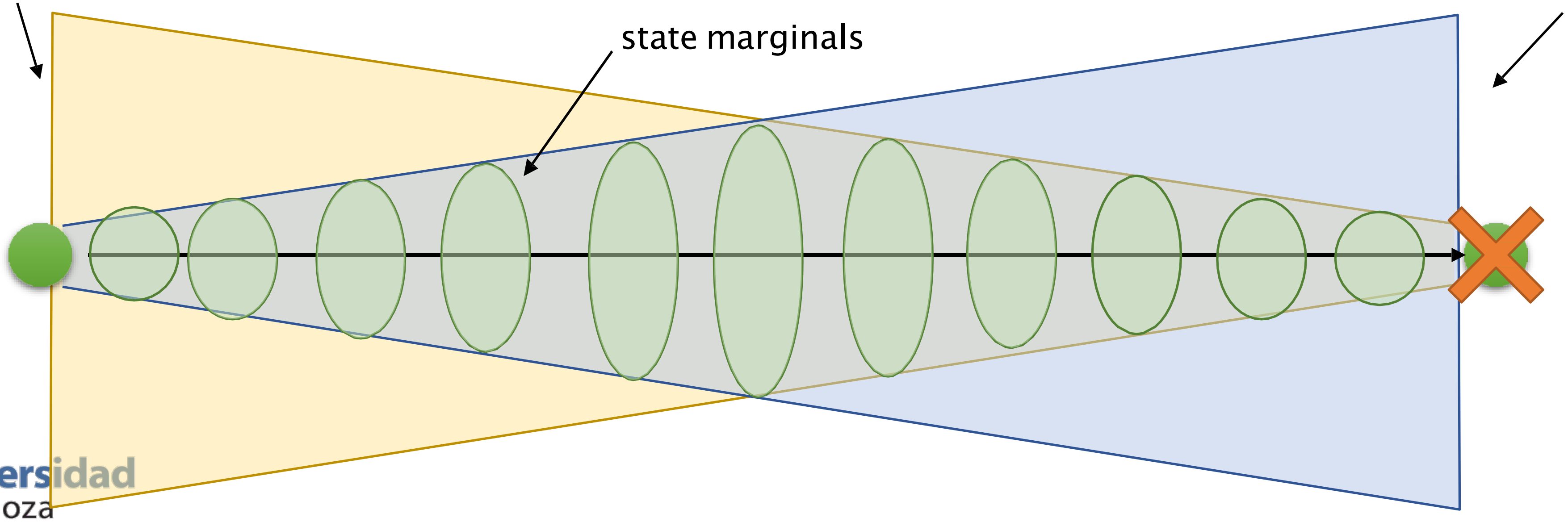


Li & Todorov, 2006

states with high probability of reaching goal

states with high probability of being reached from initial state (with high reward)

$$p(\mathbf{s}_t) \propto \beta_t(\mathbf{s}_t)\alpha_t(\mathbf{s}_t)$$



Universidad Zaragoza 1542

The optimism problem

$$Q_t(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \overbrace{\log E[\exp}^{\text{“soft-max” transition (problem!)}}(V_{t+1}(\mathbf{s}_{t+1}))]$$

$$V_t(\mathbf{s}_t) = \underbrace{\log \int}_{\text{“soft-max” action (great!)}} \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t)) d\mathbf{a}_t$$

the inference problem: $p(\mathbf{s}_{1:T}, \mathbf{a}_{1:T} | \mathcal{O}_{1:T})$

marginalizing and conditioning, we get: $p(\mathbf{a}_t | \mathbf{s}_t, \mathcal{O}_{1:T})$ (the policy)

“given that you obtained high reward, what was your action probability?”

marginalizing and conditioning, we get: $p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t, \mathcal{O}_{1:T}) \neq p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$

“given that you obtained high reward, what was your transition probability?”

Addressing the optimism problem

marginalizing and conditioning, we get: $p(\mathbf{a}_t|\mathbf{s}_t, \mathcal{O}_{1:T})$ (the policy) ⟵ **we want this**

"given that you obtained high reward, what was your action probability?"

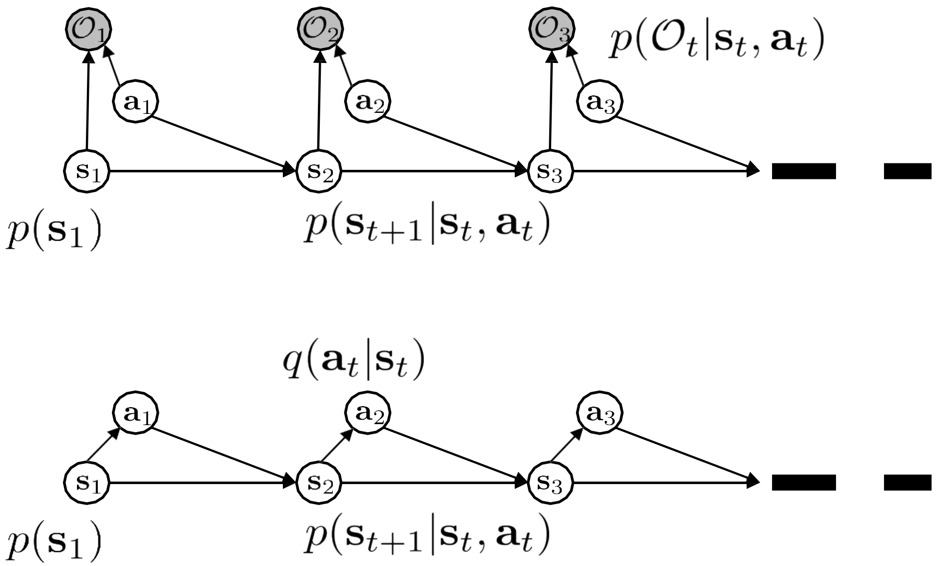
marginalizing and conditioning, we get: $p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t, \mathcal{O}_{1:T}) \neq p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$ ⟵ **but not this!**

"given that you obtained high reward, what was your transition probability?"

can we find another distribution $q(\mathbf{s}_{1:T}, \mathbf{a}_{1:T})$ that is close to $p(\mathbf{s}_{1:T}, \mathbf{a}_{1:T}|\mathcal{O}_{1:T})$ but has dynamics $p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$

$p(\mathbf{s}_{1:T}, \mathbf{a}_{1:T}|\mathcal{O}_{1:T})$

$q(\mathbf{s}_{1:T}, \mathbf{a}_{1:T})$



Variational inference!

Variational inference primer

- We want to work with intractable distribution p(x)
  - For example a posterior distribution p(x|y) or a latent variable model p(z|x)

- Let's find a *similar* tractable distribution q(x)
  - For example, a Gaussian

- How do we find similar distributions? **KL-divergence**
  - Shall we minimize $KL(p||q)$ or $KL(q||p)$?

This is intractable

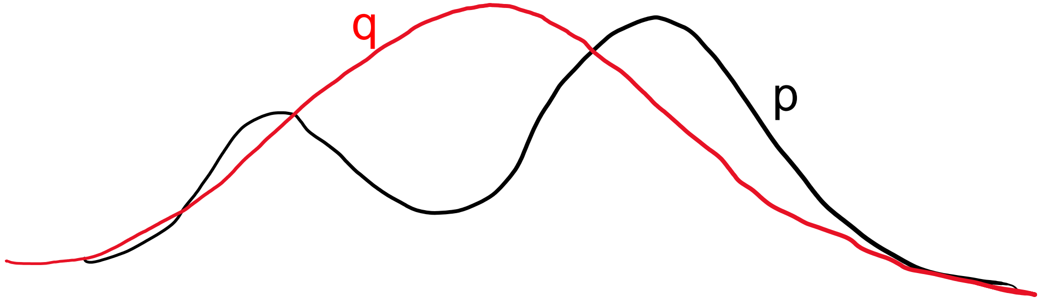
$$KL(p||q) = \int_x p(x) \log \frac{p(x)}{q(x)} dx$$

$$KL(q||p) = \int_x q(x) \log \frac{q(x)}{p(x)} dx$$

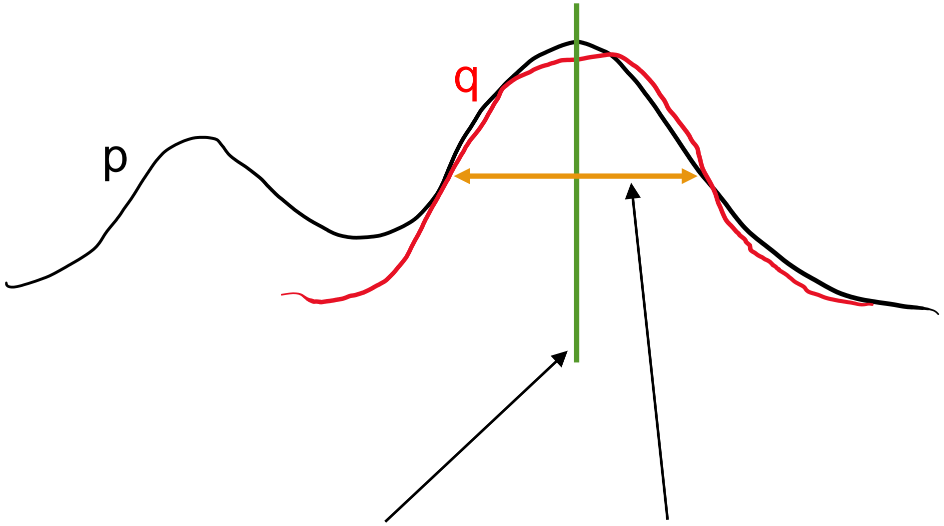
Universidad Zaragoza 1542

Variational free energy

$$KL(p||q) = \int_x p(x) \log \frac{p(x)}{q(x)} dx$$



$$KL(q||p) = \int_x q(x) \log \frac{q(x)}{p(x)} dx$$



$$KL(q||p) = \mathbb{E}_q[\log q(x)] - \mathbb{E}_q[\log p(x)] = -\mathbb{E}_q[\log p(x)] - \mathcal{H}(q(x))$$

- Minimizing KL-divergence maximizes expected log p and q entropy

Variational Bayes (this will come handy later, too)

- For Bayesian models, KL is still problematic!

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} = \frac{p(y,x)}{p(y)}$$

$$\begin{aligned} KL(q(x)||p(x|y)) &= \mathbb{E}_q[\log q(x)] - \mathbb{E}_q[\log p(x|y)] \\ &= \mathbb{E}_q[\log q(x)] - \mathbb{E}_q[\log p(x,y) - \log p(y)] \\ &= \mathbb{E}_q[\log q(x) - \log p(y,x)] + \underline{\log p(y)} \end{aligned}$$

does not depend on q

- The evidence is an intractable integral

$$p(y) = \int_x p(y|x)p(x)dx = \int_x p(y,x)dx$$

Variational Bayes with ELBO

- Let's reorganize terms:

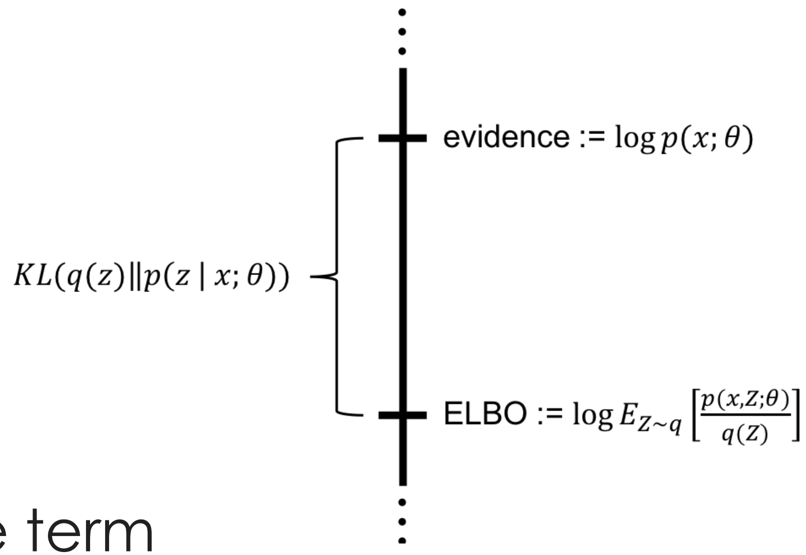
$$KL(q(x)||p(x|y)) = \mathbb{E}_q[\log q(x) - \log p(y,x)] + \log p(y)$$

- Evidence lower bound (ELBO):

$$\mathbb{E}_q[\log p(x,y) - \log q(x)] = \log p(y) - KL(q(x)||p(x|y))$$

- The left hand side is tractable!
- We can maximize the ELBO:
  - When we maximize the ELBO, we maximize the evidence/likelihood!
  - When we maximize the ELBO, we minimize the KL divergence!

Evidence Lower BOund (ELBO)



- We know that KL is a divergence:

$$KL(q||p) \geq 0$$

- Thus, the ELBO term is a lower bound of the evidence term

$$\log p(y) \geq \mathbb{E}_q[\log p(x,y) - \log q(x)]$$

- **Exercise:**
  - Using basic probability rules (Bayes, conditional, expectation, etc.) show that the ELBO is a lower bound of the evidence. You will need to use Jensen's inequality

$$\log E[y] \geq E[\log y]$$

Control via variational inference

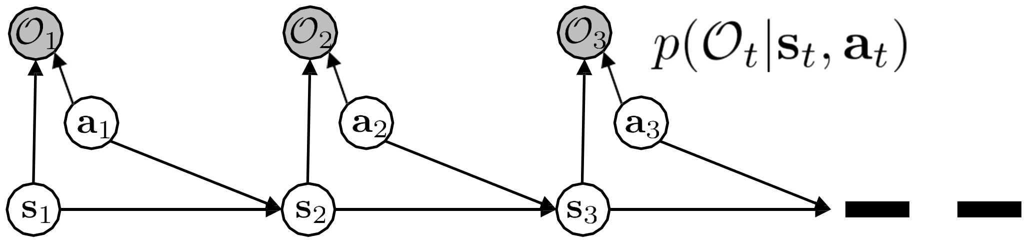
let $q(\mathbf{s}_{1:T}, \mathbf{a}_{1:T}) = p(\mathbf{s}_1) \prod_t p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) q(\mathbf{a}_t|\mathbf{s}_t)$

same dynamics and initial state as p

only new thing

let $y = \mathcal{O}_{1:T}$ and $x = (s_{1:T}, a_{1:T})$

$p(\mathbf{s}_{1:T}, \mathbf{a}_{1:T}|\mathcal{O}_{1:T})$



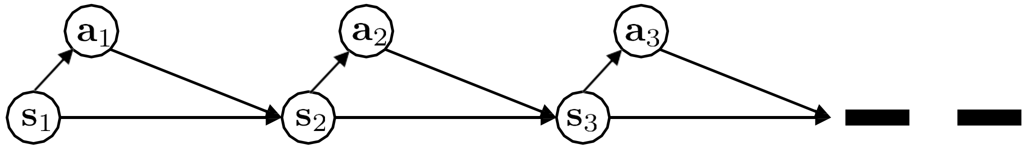
$p(\mathcal{O}_t|\mathbf{s}_t, \mathbf{a}_t)$

$p(\mathbf{s}_1)$ $p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$

$p(\mathbf{z}|\mathbf{x})$

$q(\mathbf{s}_{1:T}, \mathbf{a}_{1:T})$

$q(\mathbf{a}_t|\mathbf{s}_t)$



$p(\mathbf{s}_1)$ $p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$

$q(\mathbf{z})$

Universidad Zaragoza 1542

The variational lower bound

$$\log p(y) \geq \mathbb{E}_q[\log p(x,y) - \log q(x)]$$

let $y = \mathcal{O}_{1:T}$ and $x = (s_{1:T}, a_{1:T})$

the entropy $\mathcal{H}(q)$

let $q(\mathbf{s}_{1:T}, \mathbf{a}_{1:T}) = \underline{p(\mathbf{s}_1)}\ \underline{\prod_t p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)}q(\mathbf{a}_t|\mathbf{s}_t)$

$$\log p(\mathcal{O}_{1:T}) \geq E_{(\mathbf{s}_{1:T},\mathbf{a}_{1:T})\sim q}\Big[\cancel{\log p(\mathbf{s}_1)} + \sum_{t=1}^{T} \cancel{\log p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)} + \sum_{t=1}^{T} \log p(\mathcal{O}_t|\mathbf{s}_t, \mathbf{a}_t)$$

$$\underline{\cancel{- \log p(\mathbf{s}_1)}} - \underline{\sum_{t=1}^{T} \cancel{\log p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)}} - \sum_{t=1}^{T} \log q(\mathbf{a}_t|\mathbf{s}_t)\Big]$$

$$= E_{(\mathbf{s}_{1:T},\mathbf{a}_{1:T})\sim q}\left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t) - \log q(\mathbf{a}_t|\mathbf{s}_t)\right]$$

$$= \sum_t E_{(\mathbf{s}_t,\mathbf{a}_t)\sim q}\left[r(\mathbf{s}_t, \mathbf{a}_t) + \mathcal{H}(q(\mathbf{a}_t|\mathbf{s}_t))\right]$$

maximize reward and maximize action entropy!

Optimizing the variational lower bound

let $q(\mathbf{s}_{1:T}, \mathbf{a}_{1:T}) = p(\mathbf{s}_1) \prod_t p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) q(\mathbf{a}_t|\mathbf{s}_t)$

$$\log p(\mathcal{O}_{1:T}) \geq \sum_t E_{(\mathbf{s}_t, \mathbf{a}_t) \sim q} \left[r(\mathbf{s}_t, \mathbf{a}_t) + \mathcal{H}(q(\mathbf{a}_t|\mathbf{s}_t)) \right]$$

$$\boxed{p(x) = \arg\max \mathbb{E}_{p(x)}[g(x) - \log p(x)] \Rightarrow p(x) \propto \exp(g(x))}$$

- Use dynamic programming:

$$\begin{aligned} q(\mathbf{a}_T|\mathbf{s}_T) &= \arg\max E_{\mathbf{s}_T \sim q(\mathbf{s}_T)} \left[E_{\mathbf{a}_T \sim q(\mathbf{a}_T|\mathbf{s}_T)}[r(\mathbf{s}_T, \mathbf{a}_T)] + \mathcal{H}(q(\mathbf{a}_T|\mathbf{s}_T)) \right] \\ &= \arg\max E_{\mathbf{s}_T \sim q(\mathbf{s}_T)} \left[E_{\mathbf{a}_T \sim q(\mathbf{a}_T|\mathbf{s}_T)}[r(\mathbf{s}_T, \mathbf{a}_T) - \log q(\mathbf{a}_T|\mathbf{s}_T)] \right] \end{aligned}$$

$$q(\mathbf{a}_T|\mathbf{s}_T) = \frac{\exp(r(\mathbf{s}_T, \mathbf{a}_T))}{\int \exp(r(\mathbf{s}_T, \mathbf{a}))d\mathbf{a}} = \exp(Q(\mathbf{s}_T, \mathbf{a}_T) - V(\mathbf{s}_T))$$

$$V(\mathbf{s}_T) = \log \int \exp(Q(\mathbf{s}_T, \mathbf{a}_T)) d\mathbf{a}_T$$

Optimizing the variational lower bound

let $q(\mathbf{s}_{1:T}, \mathbf{a}_{1:T}) = p(\mathbf{s}_1) \prod_t p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) q(\mathbf{a}_t|\mathbf{s}_t)$

$$\log p(\mathcal{O}_{1:T}) \geq \sum_t E_{(\mathbf{s}_t, \mathbf{a}_t) \sim q} [r(\mathbf{s}_t, \mathbf{a}_t) + \mathcal{H}(q(\mathbf{a}_t|\mathbf{s}_t))]$$

$$\boxed{p(x) = \arg\max \mathbb{E}_{p(x)}[g(x) - \log p(x)] \Rightarrow p(x) \propto \exp(g(x))}$$

- Use dynamic programming:

Correct dynamics!

$$q(\mathbf{a}_t|\mathbf{s}_t) = \arg\max E_{\mathbf{s}_t \sim q(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim q(\mathbf{a}_t|\mathbf{s}_t)} [r(\mathbf{s}_t, \mathbf{a}_t) + E_{\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)} [V(\mathbf{s}_{t+1})]] + \mathcal{H}(q(\mathbf{a}_t|\mathbf{s}_t)) \right]$$

$$= \arg\max E_{\mathbf{s}_t \sim q(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim q(\mathbf{a}_t|\mathbf{s}_t)} [Q(\mathbf{s}_t, \mathbf{a}_t) - \log q(\mathbf{a}_t|\mathbf{s}_t)] \right]$$

Regular Bellman update!

$$q(\mathbf{a}_t|\mathbf{s}_t) = \exp(Q(\mathbf{s}_t, \mathbf{a}_t) - V(\mathbf{s}_t))$$

$$Q_t(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + E[(V_{t+1}(\mathbf{s}_{t+1})]$$

$$V_t(\mathbf{s}_t) = \log \int \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t)) d\mathbf{a}_t$$

Variational policy computation with value functions

for $t = T-1$ to 1:

$$Q_t(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + E[(V_{t+1}(\mathbf{s}_{t+1})]$$

$$V_t(\mathbf{s}_t) = \log \int \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t))\mathbf{a}_t$$

$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t) - V_t(\mathbf{s}_t)) = \exp(A_t(\mathbf{s}_t, \mathbf{a}_t))$$

- Compare:

value iteration algorithm:

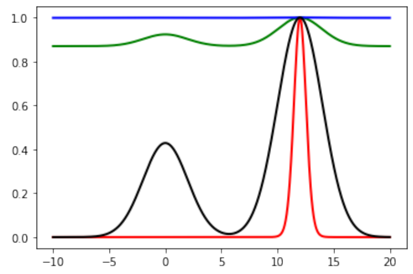
1. set $Q(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E[V(\mathbf{s}')]$
2. set $V(\mathbf{s}) \leftarrow \max_{\mathbf{a}} Q(\mathbf{s}, \mathbf{a})$

soft value iteration algorithm:

1. set $Q(\mathbf{s}, \mathbf{a}) \leftarrow r(\mathbf{s}, \mathbf{a}) + \gamma E[V(\mathbf{s}')]$
2. set $V(\mathbf{s}) \leftarrow \text{soft max}_{\mathbf{a}} Q(\mathbf{s}, \mathbf{a})$

- Variant:

explicit temperature: $V_t(\mathbf{s}_t) = \alpha \log \int \exp\left(\frac{1}{\alpha} Q_t(\mathbf{s}_t, \mathbf{a}_t)\right) d\mathbf{a}_t$



Q-learning with soft optimality

1. take some action $\mathbf{a}_i$ and observe $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i, r_i)$, add it to $\mathcal{R}$
2. sample mini-batch $\{\mathbf{s}_j, \mathbf{a}_j, \mathbf{s}'_j, r_j\}$ from $\mathcal{R}$ uniformly
3. compute $y_j = r_j + \gamma \text{soft}\max_{\mathbf{a}'_j} Q_{\phi'}(\mathbf{s}'_j, \mathbf{a}'_j)$ using *target* network $Q_{\phi'}$
4. $\phi \leftarrow \phi - \alpha \sum_j \frac{dQ_\phi}{d\phi}(\mathbf{s}_j, \mathbf{a}_j)(Q_\phi(\mathbf{s}_j, \mathbf{a}_j) - y_j)$
5. update ϕ': copy ϕ every N steps, or Polyak average $\phi' \leftarrow \tau\phi' + (1-\tau)\phi$

$$\pi(\mathbf{a}|\mathbf{s}) = \exp(Q_\phi(\mathbf{s}, \mathbf{a}) - V(\mathbf{s})) = \exp(A(\mathbf{s}, \mathbf{a}))$$

Policy gradient with soft optimality

$$\pi(\mathbf{a}|\mathbf{s}) = \exp(Q_\phi(\mathbf{s},\mathbf{a}) - V(\mathbf{s})) \text{ optimizes } \sum_t E_{\pi(\mathbf{s}_t,\mathbf{a}_t)}[r(\mathbf{s}_t,\mathbf{a}_t)] + \underbrace{E_{\pi(\mathbf{s}_t)}[\mathcal{H}(\pi(\mathbf{a}_t|\mathbf{s}_t))]}_{\text{policy entropy}}$$

- Why?

$$\pi(\mathbf{a}|\mathbf{s}) \propto \exp(Q_\phi(\mathbf{s},\mathbf{a})) \text{ when } \pi \text{ minimizes } D_{\mathrm{KL}}(\pi(\mathbf{a}|\mathbf{s}) \| \tfrac{1}{Z}\exp(Q(\mathbf{s},\mathbf{a})))$$

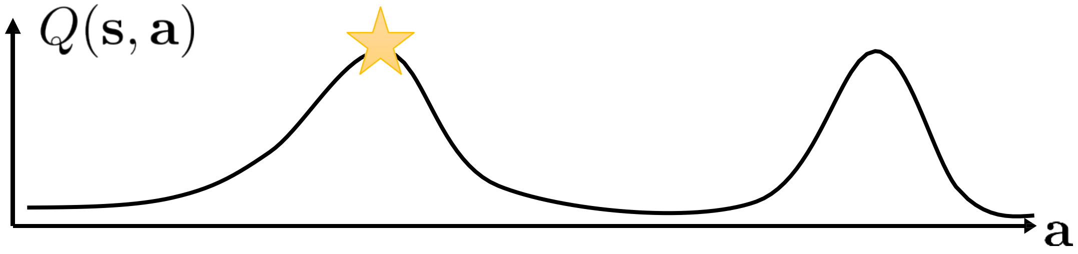
$$D_{KL}(\pi(a|s)||\tfrac{1}{Z}\exp(Q(s,a))) = -\mathbb{E}_{\pi(a|s)}[Q(s,a)] - \mathcal{H}(\pi)$$

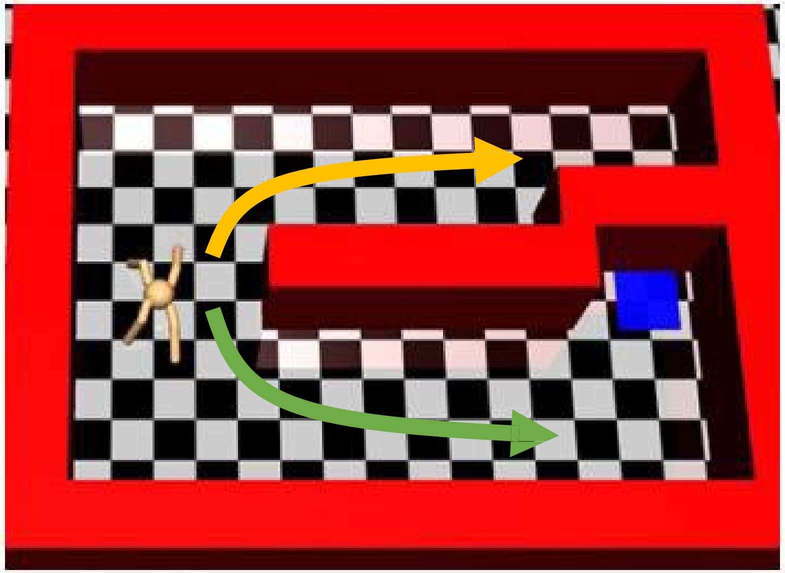
- Often referred to as “entropy regularized” policy gradient
- Policy search update lose information
- This combats premature entropy collapse

- This policy search is equivalent to soft Q-learning:
  - Published multiple times between 2017-2018!

Stochastic energy-based policies

Q-function: $Q(\mathbf{s}, \mathbf{a}) : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$



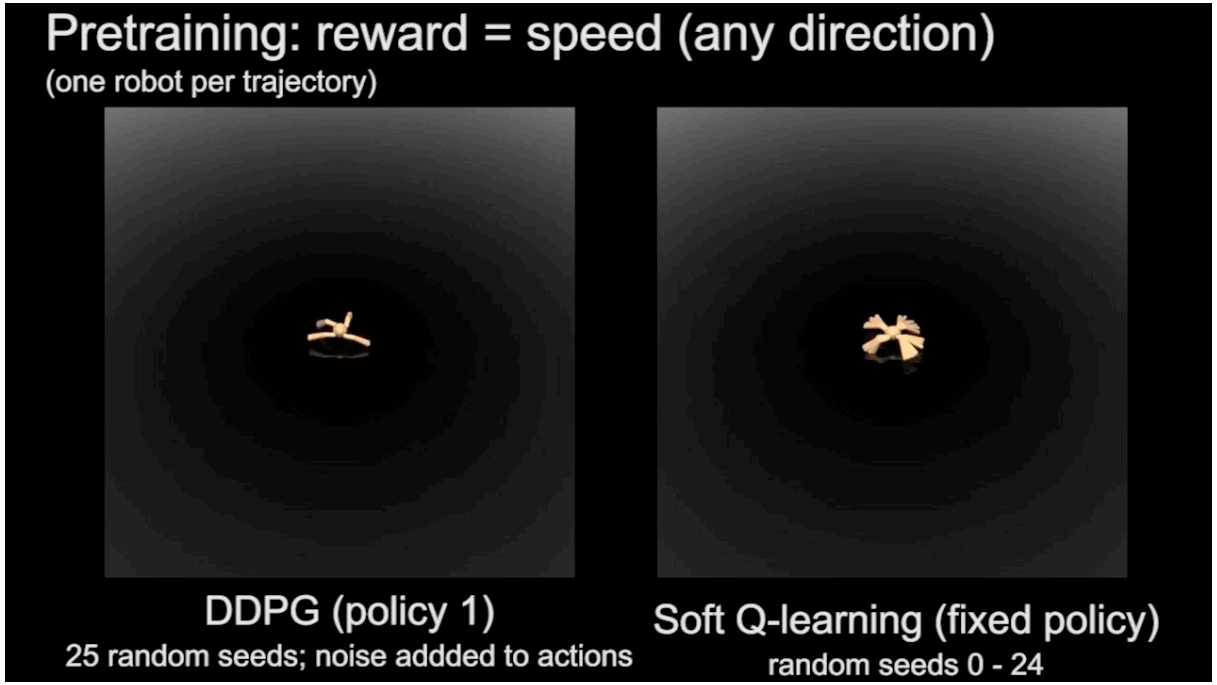


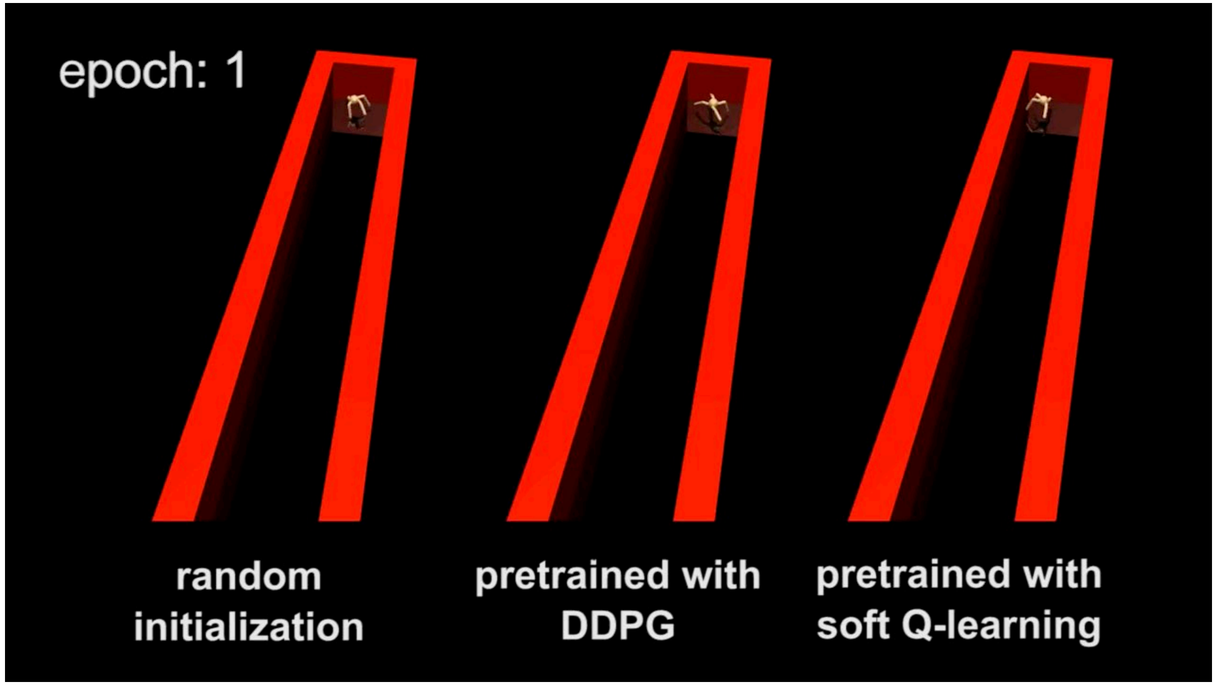
$$\pi(\mathbf{a}_t|\mathbf{s}_t) = \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t) - V_t(\mathbf{s}_t)) = \exp(A_t(\mathbf{s}_t, \mathbf{a}_t))$$

$$Q_t(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + E[V_{t+1}(\mathbf{s}_{t+1})]$$

$$V_t(\mathbf{s}_t) = \log \int \exp(Q_t(\mathbf{s}_t, \mathbf{a}_t))\mathbf{a}_t$$

Stochastic energy-based policies for pretraining





Soft actor-critic

1. Q-function update

Update Q-function to evaluate current policy:

$$Q(\mathbf{s},\mathbf{a}) \leftarrow r(\mathbf{s},\mathbf{a}) + \mathbb{E}_{\mathbf{s}'\sim p_{\mathbf{s}},\ \mathbf{a}'\sim\pi}\left[Q(\mathbf{s}',\mathbf{a}') - \log\pi(\mathbf{a}'|\mathbf{s}')\right]$$

This converges to Q^{π}.

2. Update policy

Update the policy with gradient of information projection:

$$\pi_{\text{new}} = \arg\min_{\pi'} \mathrm{D}_{\mathrm{KL}}\left(\pi'(\,\cdot\,|\mathbf{s})\,\middle\|\,\frac{1}{Z}\exp Q^{\pi_{\text{old}}}(\mathbf{s},\,\cdot\,)\right)$$

In practice, only take one gradient step on this objective

3. Interact with the world, collect more data

- One of the most popular methods right now.
- Base of many variants.
- Off-policy!
- Robust learning
- Efficient
- Adaptive

Haarnoja, Zhou, Hartikainen, Tucker, Ha, Tan, Kumar, Zhu, Gupta, Abbeel, L. Soft Actor-Critic Algorithms and Applications. ‘18

Learning to Walk via Deep Reinforcement Learning

Submission ID: 60

References

- Main reference:
  - Sergey Levine, Deep Reinforcement Learning, CS 285 at UC Berkeley
    - http://rail.eecs.berkeley.edu/deeprlcourse/
- Other courses:
  - David Silver, Reinforcement Learning, UCL
    - https://www.davidsilver.uk/teaching/
  - Nando de Freitas, Machine learning, Oxford
    - https://www.cs.ox.ac.uk/people/nando.defreitas/machinelearning/
- Books:
  - Sutton & Barto, Reinforcement Learning: An Introduction
    - http://incompleteideas.net/book/the-book-2nd.html
  - Csaba Szepesvári, Algorithms of Reinforcement Learning
    - https://sites.ualberta.ca/~szepesva/rlbook.html

Universidad Zaragoza

Applications of Deep Learning (69158)

Máster in Robotics, Graphics and Computer Vision